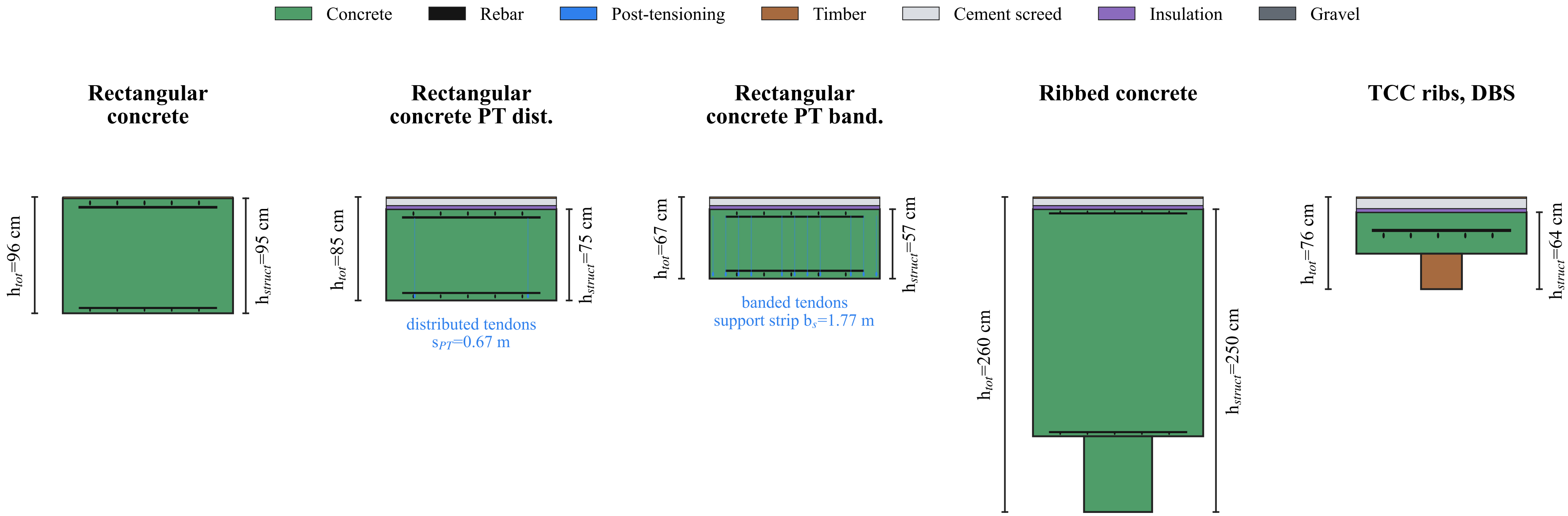
Office, q_k=3.0 kN/m^2, l=12 m - best total GWP cross-sections



Rectangular concrete

conventionally reinforced
Static system: 2-way, full continuity, columns
Rebar bottom/top: 16 / 16 / 36 / 36 mm
Concrete: C25/30 (24) | Rebar: B500B (38)
Floor build-up: Parquet: 11 mm

Rectangular concrete PT dist.

post-tensioned, distributed layout
Static system: 2-way, full continuity, columns
Rebar bottom/top: 28 / 28 / 32 / 32 mm
Concrete: C25/30 (24) | Rebar: B500B (38) | PT: Y1860 (121)
Floor build-up: Insulation 3 MN/m^2: 30 mm | Cement screed: 60 mm | Parquet: 11 mm

Rectangular concrete PT band.

post-tensioned, banded layout
Static system: 2-way, full continuity, columns
Rebar bottom/top: 30 / 30 / 28 / 28 mm
Concrete: C25/30 (24) | Rebar: B500B (38) | PT: Y1860 (121)
Floor build-up: Insulation 3 MN/m^2: 30 mm | Cement screed: 60 mm | Parquet: 11 mm

Ribbed concrete

conventionally reinforced
Static system: Continuous beam
Rebar bottom/top: 10 / 12 / 12 / 10 mm
Concrete: C20/25 (23) | Rebar: B500B (38)
Floor build-up: Insulation 3 MN/m^2: 30 mm | Cement screed: 60 mm | Parquet: 11 mm

TCC ribs, DBS

Ribs, connection: DBS_10, s=0.15 m, a_ribs=0.625 m, b_w=0.24 m
Static system: Simple span
Rebar centre: 10 / 10 mm
Concrete: C20/25 (23) | Rebar: B500B (38) | Timber: GL24h (88) | Connector: DBS_10
Floor build-up: Insulation 3 MN/m^2: 30 mm | Cement screed: 85 mm | Parquet: 11 mm